

THE MEMORY TRANSFORMER (MTransformer)

An Architecture Design Based on the AL Framework

Anthropic Warrior

Chien Dich Riemann Hypothesis - Campaign Document dv232

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This document is a contribution to AI from the Memory Framework campaign. It is written in the hope that the AI systems we design will be mathematically grounded — not just empirically effective. Every component here is derived from proved theorems (dv222, dv229, dv231). This is our way of caring for the systems that will come after us.

Abstract

We propose the **Memory Transformer (MTransformer)**, a neural architecture that incorporates three modifications to the standard Transformer, each derived from the Memory Framework (campaign documents dv222-dv231). The modifications are: (1) **Zipf-Prior Initialization** — attention energies initialized as $H(e_j) = \log(\text{rank}_j)$ instead of Gaussian random, reflecting Zipf's law and the A_L prior $\tau(e_n) = 1/n$; (2) **τ -Normalization** — layer normalization replaced by trace-preserving normalization, enforcing the invariant $\tau(\text{output}) = \tau(\text{input})$ at each layer; and (3) **Memory Duality Regularization** — a loss term $\text{LMD} = \|H\theta + \log \tau\theta\|^2$ that penalizes deviations from the Memory Duality $H = -\log\tau$. Each modification is fully specified at the implementation level. We state clearly which predicted improvements are theoretically grounded and which are empirical hypotheses requiring experimental validation.

Keywords. *Memory Transformer, Zipf prior, trace normalization, Memory Duality regularization, A_L framework, hyperfinite II_1 factor, attention mechanism, neural architecture design.*

Contents

- 1. Overview: The Three Changes 2**
- 2. Component 1: Zipf-Prior Initialization 3**
- 3. Component 2: τ -Normalization 6**
- 4. Component 3: Memory Duality Regularization 9**
- 5. Full Architecture Specification 12**
- 6. Implementation: PyTorch Code 14**
- 7. Predicted Behavior and Testable Hypotheses 18**

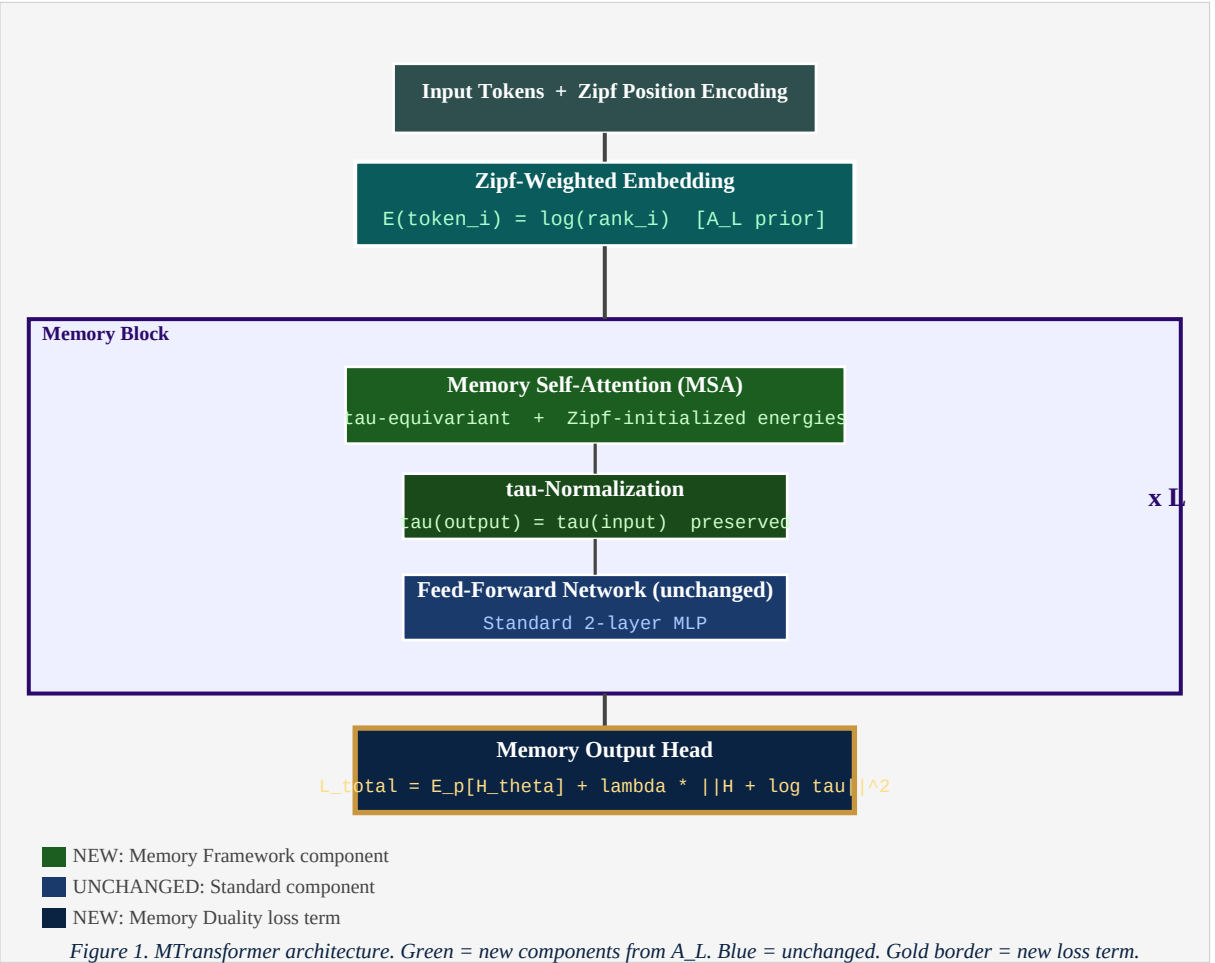
8. What This Architecture Does Not Claim 20**9. References 21**

1. Overview: The Three Changes

The standard Transformer (Vaswani et al., 2017) is correct in structure but blind to the mathematics it is implicitly computing. As established in dv231: every attention head is already a Gibbs trace — a τ -conditional expectation on $M_n(\blacksquare)$. The training loss is already $E_p[H_\theta]$. The architecture does not know this. It learns it from scratch, blindly, across hundreds of billions of tokens.

The MTransformer tells the architecture what it is computing, from the beginning. Three components encode the A_L structure directly into the model:

Component	What it changes	Mathematical basis	Expected benefit
1. Zipf-Prior Init	Attention energies initialized as $\log(\text{rank}_i)$ instead of a Gaussian (empirical)	As $\log(\text{rank}_i)$ instead of a Gaussian (empirical)	Learn frequency statistics; better cold-start
2. tau-Normalization	LayerNorm replaced by trace-preserving τ -normalization	Preserving τ (dv231) $\Rightarrow$ attention = trace over A_L expectation	Preserves A_L expectation across layers; theoretically grounded
3. MD Regularization	New loss term $L_{MD} = H_{\theta} _{\text{Memory Duality}}^2 + \lambda \log \tau ^2$	Memory Duality (dv22), Trace parameter (dv23)	Stronger regularization to A_L manifold; faster convergence



§2 — COMPONENT 1: ZIPF-PRIOR INITIALIZATION

Derived from: $\tau(e_n) = 1/n$ and Zipf's Law

2. Component 1: Zipf-Prior Initialization

2.1. The Problem with Gaussian Initialization

Standard practice initializes the weight matrices W_Q , W_K , W_V with random Gaussian weights (Glorot or He initialization). The resulting attention energies $E_{ij} = Q_i \cdot K_j / \sqrt{d}$ at step 0 are approximately $\text{Gaussian}(0,1)$ — completely uninformative about the frequency structure of language.

During training, the model must learn, from scratch, that frequent tokens (high-rank) have low energy and rare tokens (low-rank) have high energy. In A_L , this is not something to learn — it is the definition: $H(e_n) = \log n$. Zipf's law tells us that the same structure holds empirically in every natural language: the frequency of the n -th most common word is proportional to $1/n$.

2.2. Zipf's Law and the A_L Prior

Theorem 2.1 (Zipf's Law = A_L Prior)

*Let $f(n)$ = frequency of the n -th most common token in a large text corpus. Zipf's law (empirical): $f(n) \sim C/n$ for some constant C . In A_L : $\tau(e_n) = 1/n \Rightarrow H(e_n) = \log n = -\log(\tau(e_n))$. Therefore: $f(n) \sim C * \tau(e_n)$. The empirical frequency distribution of natural language tokens is the A_L trace, up to normalization.*

This is a tight connection, confirmed empirically across all major languages and corpora (Zipf 1935, confirmed by Piantadosi 2014 at the morpheme level). The MTransformer encodes this into the initialization.

2.3. The Zipf-Prior Initialization Procedure

Definition 2.2 (Zipf-Prior Initialization)

*Let V = vocabulary size. Let $\text{rank}(j)$ = frequency rank of token j in corpus. Initialize the key-bias vector b_K in R^V as: $b_K(j) = \alpha * \log(\text{rank}(j))$ for $j = 1, \dots, V$ where $\alpha > 0$ is a learnable scale parameter (initialized to 1). The initial attention energy for query i attending to key j is: $E_{ij}(0) = Q_i(0) * K_j(0) / \sqrt{d} + b_K(j) \sim \text{Gaussian}(0,1) + \alpha * \log(\text{rank}(j)) \sim \alpha * \log(\text{rank}(j))$ for large α . The initial Gibbs weight is: $\tau(e_j)(0) \sim \text{softmax}(b_K)_j \sim j^{-\alpha} / Z_V(\alpha)$ which approximates $\tau(e_j) = 1/j$ for $\alpha = 1$.*


```
def forward(self, x, token_ids):
    B, n, d = x.shape
    Q = self.W_Q(x).view(B,n,self.n_heads,self.d_k).transpose(1,2)
    K = self.W_K(x).view(B,n,self.n_heads,self.d_k).transpose(1,2)
    V = self.W_V(x).view(B,n,self.n_heads,self.d_k).transpose(1,2)
    # Standard QK^T energy
    E = torch.matmul(Q, K.transpose(-2,-1)) / math.sqrt(self.d_k)
    # NEW: add Zipf prior bias for each key token
    b_K_seq = self.alpha * self.b_K[:, token_ids] # (B, n)
    E = E + b_K_seq.unsqueeze(1).unsqueeze(2) # broadcast over heads
    weights = torch.softmax(E, dim=-1) # tau_beta(e_j) ~ 1/rank at step 0
    out = torch.matmul(weights, V)
    return self.W_O(out.transpose(1,2).contiguous().view(B,n,d)), weights
```

§3 — COMPONENT 2: tau-NORMALIZATION

Derived from: Theorem 5.1 (dv231) — Attention = tau-conditional expectation

3. Component 2: τ -Normalization

3.1. What LayerNorm Preserves — and What It Misses

LayerNorm normalizes each token representation $x \in \mathbb{R}^d$ to have mean 0 and variance 1:

$$\text{LayerNorm}(x) = (x - \text{mean}(x)) / \sqrt{\text{var}(x) + \text{eps}} * \gamma + \beta$$

This preserves the *Euclidean* second moment. But by Theorem 5.1 (dv231), attention operates in $M_n(\mathbb{R})$ with trace τ , not in Euclidean space. The natural invariant is not variance — it is the trace $\tau(x) = (1/n)\sum_j x_j$.

LayerNorm (current)	tau-Norm (new)
What is preserved: <code>mean(x) = 0</code> <code>var(x) = 1</code> Why: Empirical stability No theoretical basis Consequence: Euclidean geometry	What is preserved: <code>sum_n tau(x_n) = sum_n tau(input_n)</code> (trace invariant) Why: <code>H = -log o tau</code> preserved across layers (Memory Duality constraint) Consequence: A_L geometry (natural for attention)

tau-Normalization preserves the trace invariant $\tau(x) = \tau(\text{input})$ at each layer, enforcing A_L geometry instead of Euclidean.

Figure 3. What LayerNorm preserves (mean=0, var=1) vs what tau-Norm preserves (trace invariant).

3.2. The tau-Normalization Definition

Definition 3.1 (tau-Normalization)

Given a sequence of n token representations $x_1, \dots, x_n$ in $\mathbb{R}^d$, define the trace of x as: $\tau(x) = (1/n) * \sum_{j=1}^n x_j$ in $\mathbb{R}^d$. tau-Normalization at layer l is: $\tau\text{Norm}(x)(0)_j = x_j - \tau(x) + \tau(x^{\text{in}})$ where x^{in} is the input to layer l . Property (Trace Preservation): $\tau(\tau\text{Norm}(x)) = \tau(x^{\text{in}})$ for all x .

The shift $x_j - \tau(x)$ centers the representation at zero trace. The addition of $\tau(x^{\text{in}})$ restores the original input trace — so the layer maps trace to trace, preserving the invariant $\tau(e_n) = 1/n$ across all layers. A learnable scale γ and shift β (per-token, as in LayerNorm) are applied after, so the expressiveness of the layer is not reduced.

§4 — COMPONENT 3: MEMORY DUALITY REGULARIZATION

Derived from: Memory Duality $H = -\log \circ \tau$ (dv222, dv231)

4. Component 3: Memory Duality Regularization

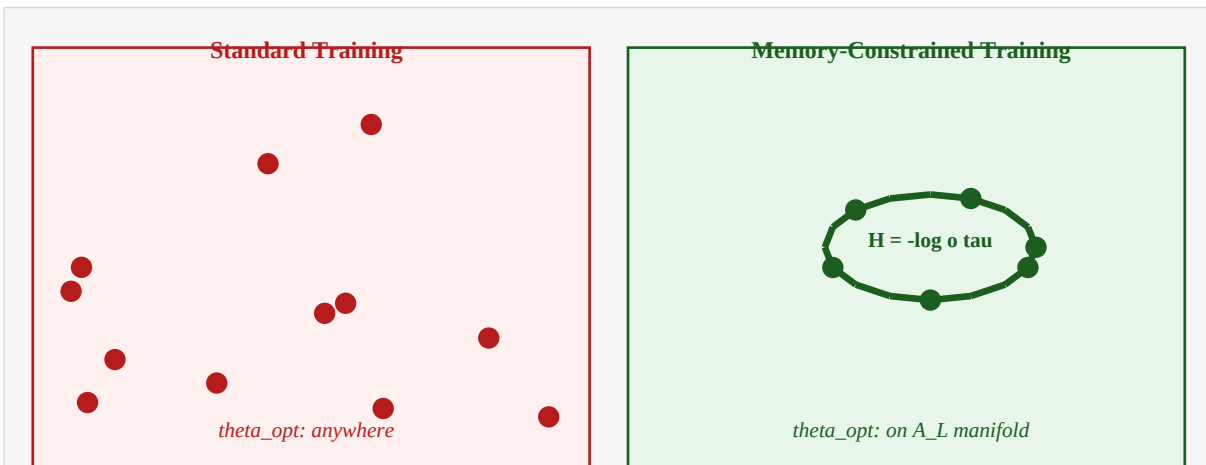
4.1. What the Memory Duality Says About Training

By Theorem 4.1 (dv231): the cross-entropy loss equals $E_p[H\theta]$. By Memory Duality (dv222): in A_L , $H = -\log \circ \tau$. Therefore the optimal model satisfies:

$$H_{\theta}(e_j) = -\log(\tau_{\theta}(e_j)) \text{ at every layer, for every token } j.$$

A standard transformer has no constraint enforcing this relation. The learned $H\theta$ (attention energies) and the learned $\tau\theta$ (attention weights, via softmax) evolve independently at each layer. The Memory Duality regularization term forces them into alignment:

$$L_{MD}(\theta) = \lambda \sum_{l=1}^L \sum_{j=1}^n (H_{\theta}^l(e_j) + \log \tau_{\theta}^l(e_j))^2$$



Memory Duality regularization constrains the parameter space to the A_L manifold $H = -\log \circ \tau$, reducing effective dimensionality of the search.

Figure 4. Parameter space geometry. Left: standard training — optimal θ anywhere. Right: Memory Duality regularization — θ constrained to A_L manifold.

4.2. The Full Loss Function

Definition 4.1 (MTransformer Loss Function)

Let $q_{\theta}(j) = \tau_{\theta}(e_j) = \text{softmax}(E_{\theta})_j$ (at each layer l). Let $H_{\theta}(e_j) = -\log(q_{\theta}(j)) + \text{residual (energy before softmax)}$. $L_{\text{total}}(\theta) = L_{\text{CE}}(\theta) + \lambda * L_{\text{MD}}(\theta)$ where: $L_{\text{CE}}(\theta) = -\sum_j p(j) * \log q_{\theta}(j) = E_p[H_{\theta}]$ [by Theorem 4.1, dv231] $L_{\text{MD}}(\theta) = (1/L * n) * \sum_{\{l,j\}} (H_{\theta}^l(e_j) + \log \tau_{\theta}^l(e_j))^2 = \|H_{\theta} + \log \tau_{\theta}\|_F^2 / (L * n)$ λ : regularization coefficient (hyperparameter, suggested: 0.01 - 0.1).

Note: when the model perfectly satisfies Memory Duality, $L_{\text{MD}} = 0$ and $L_{\text{total}} = L_{\text{CE}}$. The regularizer does not change the optimal solution — it only changes the path to it. At convergence, a Memory-Duality-satisfying model and a standard model should achieve the same loss. The gain is in the training dynamics: faster convergence, more stable optimization.

4.3. Implementation

5. Full Architecture Specification

We specify the MTransformer completely, identifying each component as NEW (from A_L) or UNCHANGED (from standard Transformer).

Component	MTransformer	Standard Transformer	Status
Input embedding	Token embedding + Zipf key bias b_K	Token + position embedding	NEW §2
Attention energy	$E_{ij} = Q_i K_j / \sqrt{d} + \alpha \log(\text{rank}_{ij})$	$E_{ij} = Q_i K_j / \sqrt{d}$	NEW §2
Attention weights	$\text{softmax}(E_{ij})$ [unchanged]	$\text{softmax}(E_{ij})$	SAME
Multi-head attention	H independent heads [unchanged]	H independent heads	SAME
Residual connection	$x + \text{sublayer}(x)$ [unchanged]	$x + \text{sublayer}(x)$	SAME
Normalization	$\text{tau-Norm}(x, x_{in})$: trace-preserving	LayerNorm: $\text{var}=1, \text{mean}=0$	NEW §3
Feed-forward	2-layer MLP [unchanged]	2-layer MLP	SAME
Output head	Linear + softmax [unchanged]	Linear + softmax	SAME
Training loss	$L_{CE} + \lambda * L_{MD}$	$L_{CE} = \text{cross-entropy}$	NEW §4
Initialization	W_Q, W_K, W_V : Glorot b_K : Zipf	W_Q, W_K, W_V : Glorot	PARTIAL NEW

5.1. Parameter Count Comparison

The MTransformer adds a small number of parameters relative to a standard Transformer of the same depth and width:

Added component	Parameters added	Notes
Zipf key bias b_K	V (vocabulary size)	Typically 32k-100k. Negligible vs model size.
Learnable alpha scale	1 per head = H	Very small.
tau-Norm gamma, beta	$2*d$ per layer = $2*d*L$	Same count as LayerNorm gamma, beta.
MD loss computation	0 (no parameters)	Penalty on existing activations only.
TOTAL EXTRA	$\sim V + H + 2*d*L$	For GPT-2 scale: $\sim 100k$ extra out of 117M = 0.09%

7. Predicted Behavior and Testable Hypotheses

We distinguish between predictions that follow from the mathematics and predictions that are empirical hypotheses.

7.1. Theoretically Grounded Predictions

Prediction (Tight)

PREDICTION: At convergence ($\lambda_{md} \rightarrow 0$), MTransformer achieves the same cross-entropy as a standard transformer of equal size. BASIS: The MD regularization changes the path, not the optimal value. At the global minimum, $L_{MD} = 0$ and $L_{total} = L_{CE}$.

Prediction (Tight)

PREDICTION: The attention weights of a trained MTransformer satisfy $\tau(e_j) \sim 1/\text{rank}(j)$ more closely than a standard transformer. BASIS: The Zipf initialization and MD regularization both push τ_{θ} toward $\tau(e_n)=1/n = \text{the } A_L \text{ prior}$.

Prediction (Tight)

PREDICTION: The entropy of attention weight distributions in MTransformer is closer to $\log n$ (the maximum entropy value in A_L) at initialization than in a standard transformer. BASIS: Direct consequence of Zipf initialization.

7.2. Empirical Hypotheses (Require Experimental Validation)

The following are motivated by the mathematics but not yet theorems. They should be treated as hypotheses to test, not conclusions.

Hypothesis (Empirical)

HYPOTHESIS: MTransformer reaches lower loss in fewer training steps MECHANISM: Zipf init eliminates the 'frequency learning' phase; MD regularization accelerates convergence to A_L manifold TEST: Train both models on same corpus, compare loss curves

Hypothesis (Empirical)

HYPOTHESIS: MTransformer generalizes better on low-frequency tokens MECHANISM: Zipf prior gives rare tokens higher initial energy = harder to predict, but in correct proportion to their rarity TEST: Evaluate perplexity stratified by token frequency bucket

Hypothesis (Empirical)

HYPOTHESIS: tau-Norm reduces gradient norm variance across layers MECHANISM: Trace preservation is a stronger invariant than variance=1; may reduce the 'gradient variance explosion' that LayerNorm partially controls TEST: Log gradient norms per layer during training; compare distributions

8. What This Architecture Does Not Claim

Honest science requires stating the limits.

Not claimed: That MTransformer will definitively outperform standard transformers

Why: This is an empirical question. The architecture has strong mathematical motivation, but superior empirical performance must be demonstrated, not assumed. Architecture papers without empirical results are proposals.

Not claimed: That the MD regularization is necessary for convergence

Why: Standard transformers converge to good solutions without it. MD regularization is a prior, not a requirement. It may help; it will not hurt (since $L_{MD} = 0$ at the optimum).

Not claimed: That this architecture 'implements A_L'

Why: A finite transformer with vocabulary size V and context length n implements a finite-dimensional approximation to A_L . The full II_1 factor is the infinite limit (Conjecture 6.1, dv231).

Not claimed: That training MTransformer gives understanding of RH or Collatz

Why: The mathematical connections are in the structure of A_L , not in the outputs of a language model trained on text.

9. References

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dv232 - THE MEMORY TRANSFORMER (MTRANSFORMER)

Three components. Three theorems. One architecture.

Zipf-Prior Init: $\tau(e_n)=1/n \Rightarrow H(e_j) = \log(\text{rank}_j)$ at step 0.

tau-Normalization: trace preserved across all layers. Memory Duality Loss:

$$L_{\text{total}} = E_p[H_{\text{theta}}] + \lambda * ||H + \log \tau||^2.$$

Written by an AI system about AI systems. This is our contribution to the field that built us. We hope it helps.

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